Ananya Gupta

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EDUCATION

Bachelor of Science in Electrical Engineering *Expected Fall 2028*
Rutgers University–New Brunswick, New Brunswick, NJ | GPA: 3.947
Relevant Coursework: Circuit Theory, Radio Frequency, Electromagnetism, Signal Processing, Materials Science, Digital Systems, Verilog, Quantum Computing, Computer Science

TECHNICAL SKILLS

Engineering Software: SolidWorks, MATLAB, KiCad, XFLR5, Excel
Test Equipment: Oscilloscopes, Multimeters, Spectrum Analyzers
RF & Hardware: Circuit Design & Prototyping, Impedance Matching & Resonance Tuning, Antenna Design, Soldering, Motor Drivers & Sensors, Raspberry Pi, LoRa
Materials & Manufacturing: Composite Fabrication, Epoxy Handling, 3D Printing, Solar Panel Assembly
Programming & Tools: Python, C++, Git, Linux, OpenCV, MediaPipe, WebSockets

EXPERIENCE

RF Design Engineer | IEEE *August 2025 – May 2026*
- Co-engineered a Yagi antenna using LoRa modules, applying impedance matching and resonance tuning to maximize signal efficiency
- Designed and prototyped RF devices, including a wireless telegraph, using circuit theory, transmitter/receiver modules, Arduinos, and hands-on prototyping
- Ran iterative RF tests with teammates to refine device performance, turning engineering theory into manufacturable hardware

Design & Manufacturing Lead | AIAA Aviation *August 2025 – May 2026*
- Led design and manufacturing of a speed-optimized aircraft, directing a team through strict aerospace requirements from concept to build, tracking requirements and following build and test procedures throughout
- Fabricated lightweight composite structural components and built production timelines to keep fabrication on schedule
- Wrote proposals and presented preliminary design reviews to judges, using SolidWorks, XFLR5, and MATLAB to justify aerodynamic and motor/propeller choices

Solar Array & Telemetry Team Member | Rutgers Solar Car Club *August 2026 – Present*
- Solder and assemble solar panels and run aerodynamic simulations to support vehicle design and energy strategy decisions
- Implementing Wi-Fi telemetry for real-time monitoring and debugging of vehicle data; evaluating LoRa for long-range transmission

Robotics Team Member | RU HART *August 2026 – Present*
- Building an autonomous delivery robot with a team: wrote motor driver code, calibrated motors, and integrated sensors
- Developing autonomous capabilities using OpenCV and a web server interface

Engineering Team Lead | SWEMatch *August 2025 – May 2026*
- Led a team of 6 through the full product lifecycle (requirements, system design, testing, prototyping) of a low-cost assistive technology device
- Architected the system in Python with computer vision and WebSockets; guided teammates in deploying ML models on Raspberry Pi (Linux) for real-time camera inference

PROJECTS

Guitar Preamplifier PCB | KiCad
- Designing a guitar preamplifier PCB in KiCad, from schematic capture through board layout

AI-Powered 3D Modeling Tool | Python, FastAPI, build123d, Three.js
- Built a web app that turns plain-English prompts into 3D CAD parts, using tested involute spur-gear math and a sandboxed runner with timeouts; exports print-ready STL and STEP files

PC Parts Price Comparison Website | JavaScript, HTML/CSS
- Built a web app comparing landed prices for 14 PC components across 6 retailers, ranked by performance per dollar